

Acceptance probabilities fluctuate even under a fixed kernel. A temporal block below the practical band and another above it therefore do not by themselves show a change in the chain’s acceptance distribution. The default raw crossing screen is a heuristic, and its separate four-chain compatibility interval does not calibrate the crossing decision. Independent random seeds also do not make four differently initialized chain means identically distributed.

For a stationary acceptance sequence a_{it} with an applicable central limit theorem, uncertainty in its mean depends on the long-run variance $\sigma_i^2 = \gamma_i(0) + 2 \sum_{k \geq 1} \gamma_i(k)$, where $\gamma_i(k)$ is its lag- k autocovariance. Independence between m chains of length n then gives the approximation

$$\widehat{\text{Var}}(\bar{a}) = \frac{1}{m^2 n} \sum_{i=1}^m \hat{\sigma}_i^2.$$

This follows by adding the variances of the independent chain means and dividing by m^2 . Batch means and lugsail estimate these long-run variances; the formulas and assumptions are discussed in the diagnostics chapter. A positive estimate cannot establish adequate batch length, stationarity or sufficient burn-in.

An experimental acceptance diagnostic now preserves the original tuning decision while computing these uncertainty estimates separately. It compares all temporal windows within each chain, so opposing drifts cannot cancel by averaging across chains. It subtracts simultaneous marginal interval endpoints: if both means are covered, their difference is covered regardless of their covariance. A difference interval wholly outside a declared tolerance supports a material discrepancy; an interval wholly inside supports compatibility for that contrast; overlap with the tolerance boundary remains unresolved. The approximate marginal intervals still require calibration, including across the declared evidence looks and candidate search.

The October 1 validation did not support a new default. Under strongly persistent stationary simulations, the tested batch rules underestimated uncertainty and reported false conflicts. Increasing the batch length reduced some conflicts without establishing adequate coverage. The new diagnostic therefore has no tuning-artifact authority. Its finite-start model checks and BGS replays are separate from posterior convergence, and R-hat remains reporting-only during tuning.